SignMuon: fast as Muon, communication-effective as SignSGD

1. Centralized Problem:

$$\min_{\mathbf{x}\in\mathbb{R}^d} f(\mathbf{x}) = \mathbb{E}_{\xi\sim\mathcal{J}}[f(\mathbf{x},\xi)]$$

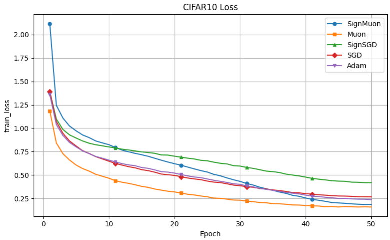
2. Federated Objective:

$$F(\mathbf{x}) = \min_{\mathbf{x}\in\mathbb{R}^d} \frac{1}{M}\sum_{m=1}^{M} \underbrace{\mathbb{E}_{\xi_m\sim D_m} f(\mathbf{x},\xi_m)}_{\text{local loss}}$$

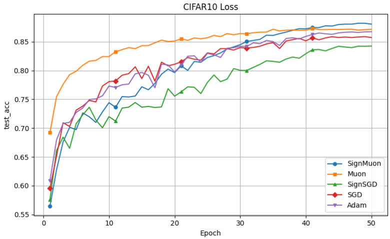
SignMuon Algorithm

1. $\mathbf{M}_t = \mu\mathbf{M}_{t-1} + \mathbf{G}_t$ (momentum)
2. $\mathbf{D}_t = \mathrm{LMO}(\mathbf{M}_t) \approx \mathbf{U}_t\mathbf{V}_t^\top$ (Newton–Schulz, 5th order)
3. $\mathbf{s}_t = \mathrm{sign}(\mathbf{D}_t)$ (1-bit compression)
4. $\mathbf{X}_{t+1} = \mathbf{X}_t - \eta\mathbf{s}_t$ (update)

Preliminary Results (CIFAR-10)



(a) Loss on train dataset



(b) Accuracy on test dataset